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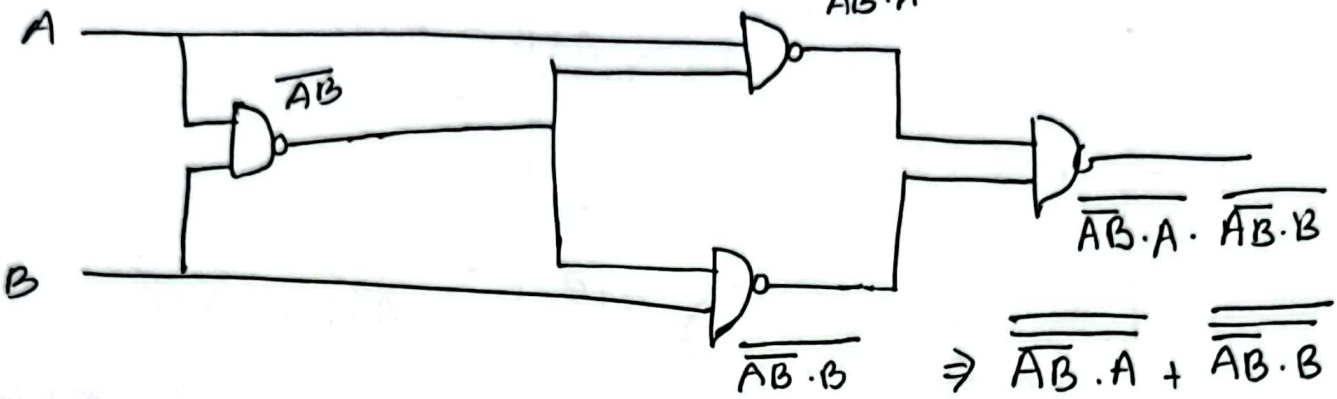
Circuit diagram 1



$$\Rightarrow \overline{\overline{\overline{AB}\cdot A}} + \overline{\overline{\overline{AB}\cdot B}}$$

$$\Rightarrow \overline{AB}\cdot A + \overline{AB}\cdot B$$

$$\Rightarrow (\bar{A}+\bar{B}).A + (\bar{A}+\bar{B}).B$$

$$\Rightarrow \bar{A}A + \bar{B}A + \bar{A}B + \bar{B}B$$

$$= \bar{A}B + A\bar{B}$$

$$= A \oplus B$$

Truth table

A	B	out
0	0	0
0	1	1
1	0	1
1	1	0

A	B	$\overline{AB}$	$\overline{\overline{AB}\cdot A}$	$\overline{\overline{AB}\cdot B}$	$A \oplus B$
0	0	1	0	0	0
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	0	0

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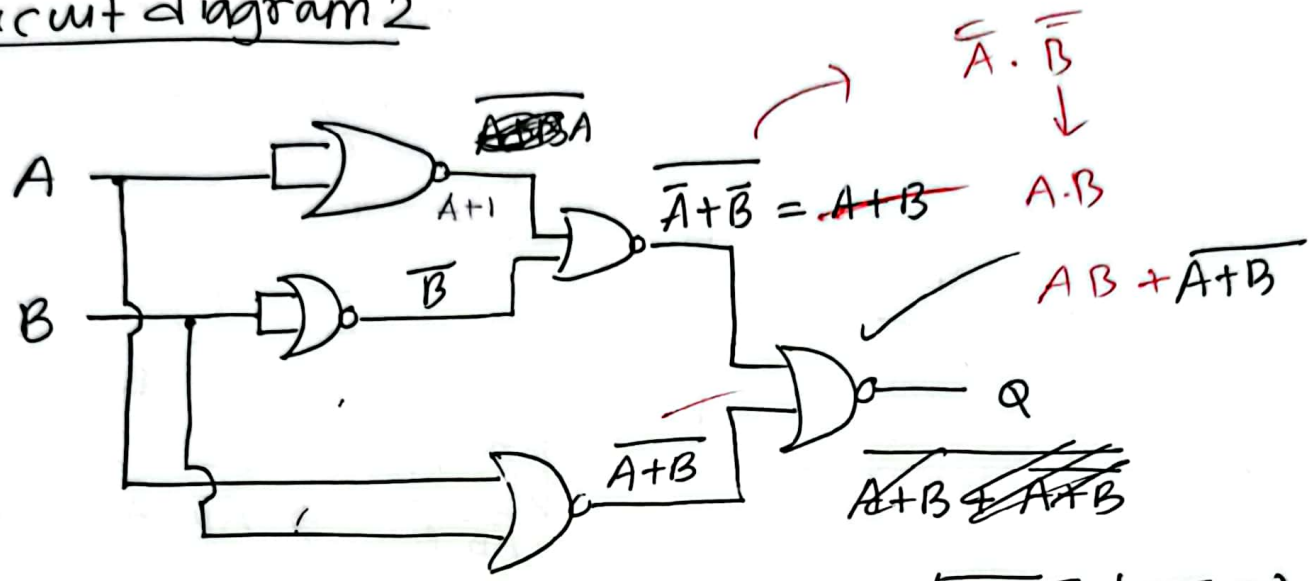
circuit diagram 2



$\cancel{\overline{A+B} + \overline{A+B}}$

$= \cancel{(\overline{A+B}) + (\overline{A+B})}$

$= \cancel{\overline{AB} + \overline{A+B}}$

$= \overline{\underbrace{AB}_{x} + \underbrace{\overline{A+B}}_{y}}$

$= \cancel{AB + \overline{A}.\overline{B}}$

$= \cancel{A+B}$

$= \overline{AB} . \overline{\overline{A+B}}$

$= (\overline{A}+\overline{B}).(A+B)$

$= \cancel{\overline{A}A} + \overline{A}B + \overline{B}A + \cancel{\overline{B}B}$

$= \overline{A}B + A\overline{B}$

$= A \oplus B$

Truth table

A	B	A+B	$\overline{A+B}$	$\overline{A+B+\overline{A+B}}$
0	0	0	1	1
0	1	1	0	1
1	0	1	0	1
1	1	1	0	0

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Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 2: Universal Gates, Applications of Boolean Algebra

Objective:

- To investigate the rules of Boolean algebra.
- To gain experience working with practical circuits.
- To simplify a complex function using Boolean algebra.

Required Components:

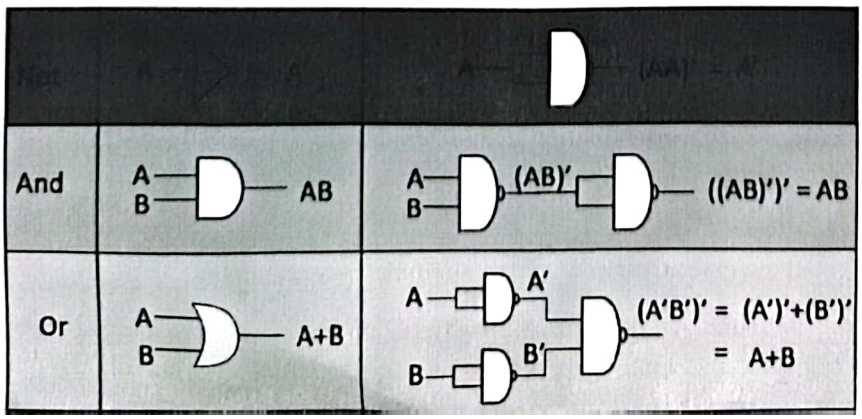
1. IC 7400 × 1
2. IC 7402 x 2

Boolean Postulates & Theorems:

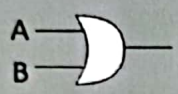
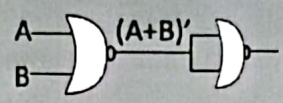
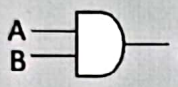
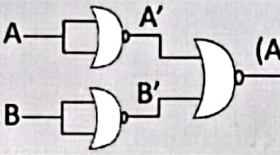
Postulate 2:	(a) $x + 0 = x$	(b) $x \cdot 1 = x$
Postulate 3: Commutative	(a) $x + y = y + x$	(b) $x \cdot y = y \cdot x$
Postulate 4: Distributive	(a) $x(y+z) = xy + xz$	(b) $x + yz = (x+y).(x+z)$
Postulate 5:	(a) $x + x' = 1$	(b) $x \cdot x' = 0$
Theorem 1:	(a) $x + x = x$	(b) $x \cdot x = x$
Theorem 2:	(a) $x + 1 = 1$	(b) $x \cdot 0 = 0$
Theorem 3: Involution	$(x')' = x$	
Theorem 4: Associative	(a) $x+(y+z) = (x+y) +z$	(b) $x.(y.z) = (x.y).z$
Theorem 5: DeMorgan	(a) $(x + y)' = x' \cdot y'$	(b) $(xy)' = x' + y'$
Theorem 6: Absorption	(a) $x + xy = x$	(b) $x. (x + y) = x$

Diagrams:

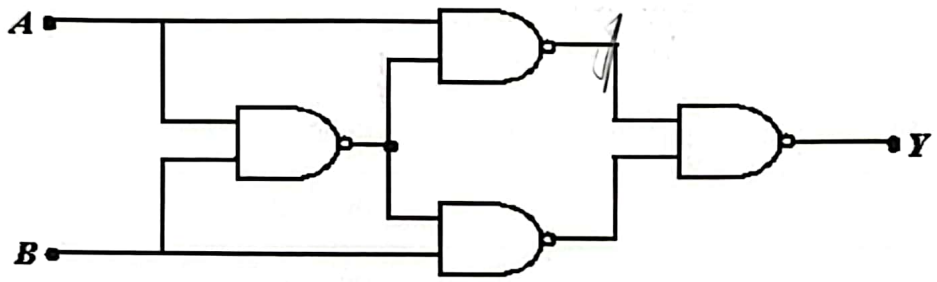
Building basic gates using Universal (NAND) Gate(s):



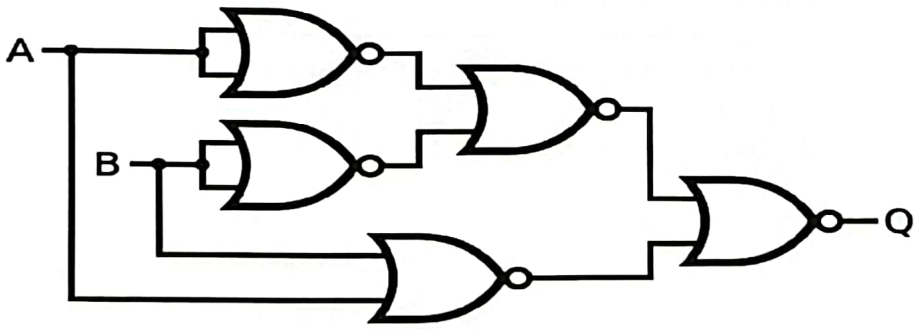
Building basic gates using Universal (NOR) Gate(s):

Or		
And		

Circuit Diagram - 1:



Circuit Diagram - 2:



Procedure:

- Construct the Circuit Diagram - 1 on the breadboard.
- Remember to connect each IC's VCC pin to the "+5V" position of the DC Power Supply of the trainer board, and the GND or 0V pin to the "GND" position of the trainer board.
- Connect the inputs to the Data switches and outputs to any position of the LED Display.
- Find out the outputs for all possible combinations of input states.
- Write down the input-output in tabular form.

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